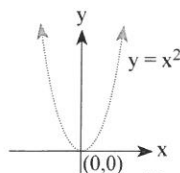


Quadratics

A quadratic can be sketched by transformations of $y = x^2$.



Sketch - no units shown.

Sketch - label the turning point.

Exercise 16.7

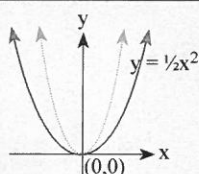
Sketch each of the following quadratics by first sketching $y = x^2$ and then applying the appropriate transformations:

ax^2 the parabola is $\cup$ for $a > 1$.

ax^2 the parabola is $\cap$ for $a < 1$.

$$y = \frac{1}{2}x^2$$

The coefficient, a , is less than 1, thus the shape is wider.



1 $y = 3x^2$

2 $y = \frac{1}{4}x^2$

3 $y = 10x^2$

4 $y = 0.75x^2$

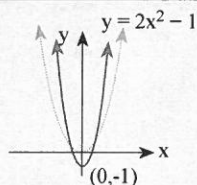
$ax^2 + d$ the parabola is shifted d units up.

$ax^2 - d$ the parabola is shifted d units down.

Use Technology to check your answers.

$$y = 2x^2 - 1$$

The coefficient = 2 thus the shape is narrower.
The parabola is also translated 1 unit down



5 $y = 2x^2 + 3$

6 $y = 3x^2 - 1$

7 $y = 0.5x^2 + 1$

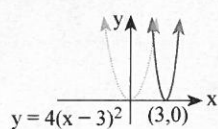
8 $y = \frac{3}{4}x^2 + 4$

$a(x + d)^2$ the parabola is shifted d units left.

$a(x^2 - d)^2$ the parabola is shifted d units right.

$$y = 3(x - 3)^2$$

The parabola is translated 3 units to the right.
The coefficient = 3 thus the shape is narrower.



9 $y = (x + 3)^2$

10 $y = (x - 1)^2$

11 $y = (x + 1)^2$

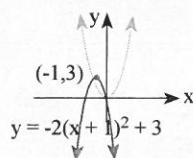
12 $y = 3(x - 4)^2$

ax^2 the parabola has a $\cup$ shape.

$-ax^2$ the parabola has a $\cap$ shape.

$$y = -2(x + 1)^2 + 3$$

The parabola is reflected vertically, is narrower, is translated horizontally 1 unit to the left and is translated vertically 3 units.



13 $y = -2(x + 1)^2 + 3$

14 $y = -3(x + 2)^2 - 1$

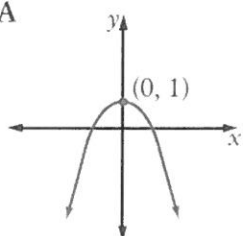
15 $y = -0.5(x - 1)^2 + 1$

16 $y = -\frac{3}{4}(x - 2)^2 + 4$

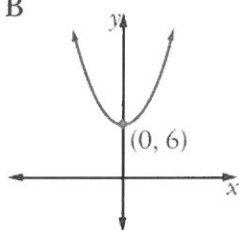
2 Match each of these equations with a parabola from those shown below.

- | | | |
|-------------------|------------------|------------------|
| a $y = 1 + x^2$ | b $y = 10 - x^2$ | c $y = x^2 - 6$ |
| d $y = -10 - x^2$ | e $y = x^2$ | f $y = 6 - x^2$ |
| g $y = x^2 + 6$ | h $y = -x^2 + 1$ | i $y = x^2 - 10$ |
| j $y = -x^2$ | k $y = -x^2 - 6$ | l $y = x^2 + 10$ |

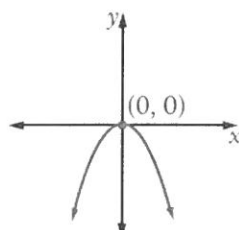
A



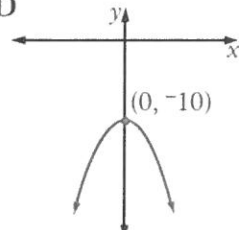
B



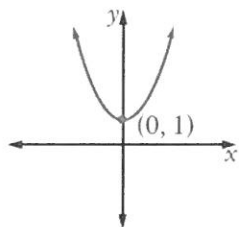
C



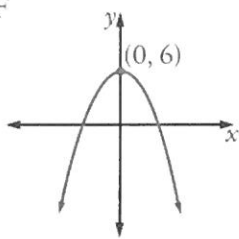
D



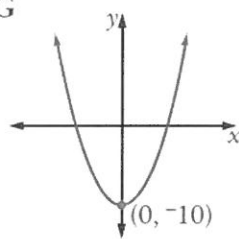
E



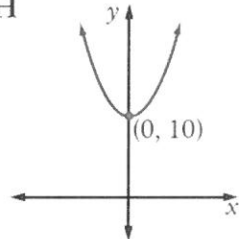
F



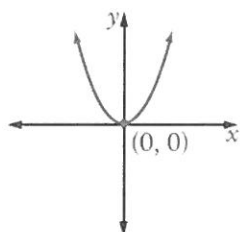
G



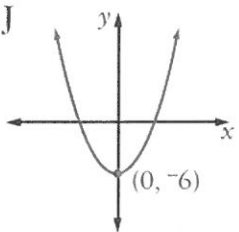
H



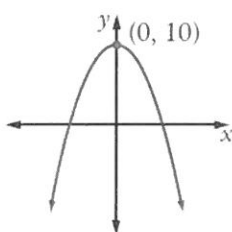
I



J



K



L

